

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	ATHIN SIVA.S	Reg. No.:	192524237
Section / Batch:	2025	Date:	28/7/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Code of Conduct

I ____ATHIN SIVA.S____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Q61. Closest Pair – Warehouse Location Analysis

Problem Interpretation

A logistics company has six warehouses located at the following coordinates:

[
{(2,1), (5,4), (6,5), (10,9), (3,2), (8,7)}
]

The objective is to find the two warehouses that are closest to each other using the **Brute Force Closest Pair Algorithm**. Identifying the nearest warehouses helps reduce transportation costs, improve inventory sharing, and increase delivery efficiency.

Justification for Using Brute Force

The Brute Force method is suitable because:

- It compares every possible pair of warehouses.
- It guarantees the correct closest pair.
- It is simple to implement.
- It works well for small datasets.

Step 1: Label the Warehouses

Warehouse Coordinates

W1	(2,1)
W2	(5,4)
W3	(6,5)
W4	(10,9)
W5	(3,2)
W6	(8,7)

Step 2: Distance Formula

The Euclidean distance between two warehouses is

[
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
]

Step 3: Compute Distances Between All Pairs

W1 – W2

$$\begin{aligned} &[\\ &\sqrt{(5-2)^2+(4-1)^2} \\ &=\sqrt{9+9} \\ &=\sqrt{18} \\ &=4.24 \\ &] \end{aligned}$$

W1 – W3

$$\begin{aligned} &[\\ &\sqrt{(6-2)^2+(5-1)^2} \\ &=\sqrt{16+16} \\ &=\sqrt{32} \\ &=5.66 \\ &] \end{aligned}$$

W1 – W4

$$\begin{aligned} &[\\ &\sqrt{(10-2)^2+(9-1)^2} \\ &=\sqrt{64+64} \\ &=\sqrt{128} \\ &=11.31 \\ &] \end{aligned}$$

W1 – W5

$$\begin{aligned} &[\\ &\sqrt{(3-2)^2+(2-1)^2} \\ &=\sqrt{1+1} \\ &=\sqrt{2} \\ &=1.41 \\ &] \end{aligned}$$

W1 – W6

$$\begin{aligned} &[\\ &\sqrt{(8-2)^2+(7-1)^2} \\ &=\sqrt{36+36} \\ &=\sqrt{72} \\ &=8.49 \\ &] \end{aligned}$$

W2 – W3

$$\begin{aligned} &[\\ &\sqrt{(6-5)^2+(5-4)^2} \end{aligned}$$

$$=\sqrt{1+1}$$

$$=\sqrt{2}$$

$$=1.41$$

$$]$$

W2 – W4

$$[$$

$$\sqrt{(10-5)^2+(9-4)^2}$$

$$=\sqrt{25+25}$$

$$=\sqrt{50}$$

$$=7.07$$

$$]$$

W2 – W5

$$[$$

$$\sqrt{(3-5)^2+(2-4)^2}$$

$$=\sqrt{4+4}$$

$$=\sqrt{8}$$

$$=2.83$$

$$]$$

W2 – W6

$$[$$

$$\sqrt{(8-5)^2+(7-4)^2}$$

$$=\sqrt{9+9}$$

$$=\sqrt{18}$$

$$=4.24$$

$$]$$

W3 – W4

$$[$$

$$\sqrt{(10-6)^2+(9-5)^2}$$

$$=\sqrt{16+16}$$

$$=\sqrt{32}$$

$$=5.66$$

$$]$$

W3 – W5

$$[$$

$$\sqrt{(3-6)^2+(2-5)^2}$$

$$=\sqrt{9+9}$$

$$=\sqrt{18}$$

$$=4.24$$

$$]$$

W3 – W6

$$\begin{aligned}
 &[\\
 &\sqrt{(8-6)^2+(7-5)^2} \\
 &=\sqrt{4+4} \\
 &=\sqrt{8} \\
 &=2.83 \\
 &]
 \end{aligned}$$

W4 – W5

$$\begin{aligned}
 &[\\
 &\sqrt{(3-10)^2+(2-9)^2} \\
 &=\sqrt{49+49} \\
 &=\sqrt{98} \\
 &=9.90 \\
 &]
 \end{aligned}$$

W4 – W6

$$\begin{aligned}
 &[\\
 &\sqrt{(8-10)^2+(7-9)^2} \\
 &=\sqrt{4+4} \\
 &=\sqrt{8} \\
 &=2.83 \\
 &]
 \end{aligned}$$

W5 – W6

$$\begin{aligned}
 &[\\
 &\sqrt{(8-3)^2+(7-2)^2} \\
 &=\sqrt{25+25} \\
 &=\sqrt{50} \\
 &=7.07 \\
 &]
 \end{aligned}$$

Step 4: Distance Table

Pair	Distance
W1 – W2	4.24
W1 – W3	5.66
W1 – W4	11.31
W1 – W5	1.41
W1 – W6	8.49
W2 – W3	1.41

Pair	Distance
------	----------

W2 – W4	7.07
---------	------

W2 – W5	2.83
---------	------

W2 – W6	4.24
---------	------

W3 – W4	5.66
---------	------

W3 – W5	4.24
---------	------

W3 – W6	2.83
---------	------

W4 – W5	9.90
---------	------

W4 – W6	2.83
---------	------

W5 – W6	7.07
---------	------

Step 5: Closest Pair

The minimum distance obtained is

[
 $\boxed{\sqrt{2}=1.41\text{ units}}$
]

The closest warehouse pairs are:

- **W1 (2,1) and W5 (3,2)**
- **W2 (5,4) and W3 (6,5)**

Both pairs have the same minimum distance of **1.41 units**.

Step 6: Number of Pairwise Comparisons

For **n = 6** warehouses,

[
 $\frac{n(n-1)}{2}$
 $\frac{6 \times 5}{2}$
15
]

Total Pairwise Comparisons = 15

Time Complexity Analysis

Case	Complexity
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Best Case	$O(n^2)$
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Average Case	$O(n^2)$
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Worst Case	$O(n^2)$
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Since every possible pair of warehouses must be examined, the algorithm always performs $O(n^2)$ comparisons regardless of the input.

Correctness Verification

- Every possible pair of warehouses was compared.
- Euclidean distance was calculated correctly for each pair.
- The smallest distance (**1.41 units**) was identified.
- Therefore, the Brute Force algorithm correctly determines the nearest warehouses.

Logistics Optimization

Finding the closest warehouses helps to:

- Reduce transportation distance and fuel consumption.
- Improve inventory sharing between nearby warehouses.
- Enable faster emergency stock transfers.
- Minimize operational costs.
- Enhance overall supply chain efficiency.

Result

The **Brute Force Closest Pair Algorithm** successfully examined all **15** warehouse pairs and found that the closest warehouses are **W1–W5** and **W2–W3**, each with a minimum distance of **1.41 units**. The algorithm guarantees the correct result with a time complexity of $O(n^2)$ and is suitable for small datasets used in logistics planning.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex realworld problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & DecisionMaking Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				

Total Marks (100):	
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Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____